

Assignment ::- 10

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Subject	DBMS

Q1) Create a MySQL database named SalesDB and a table named SalesData.

SalesData Table

Column Name	Data Type	Constraint
SaleID	INT	PRIMARY KEY
Salesperson	VARCHAR(50)	NOT NULL
Department	VARCHAR(30)	NOT NULL
SaleDate	DATE	NOT NULL
Amount	DECIMAL(10,2)	NOT NULL

Insert at least 12 sales records belonging to three different departments.

-- Step 1: Create the database

```
CREATE DATABASE SalesDB;
```

-- Step 2: Select the database

```
USE SalesDB;
```

-- Step 3: Create the SalesData table

```
CREATE TABLE SalesData (  
    SaleID INT PRIMARY KEY,  
    Salesperson VARCHAR(50) NOT NULL,  
    Department VARCHAR(30) NOT NULL,  
    SaleDate DATE NOT NULL,  
    Amount DECIMAL(10,2) NOT NULL  
);
```

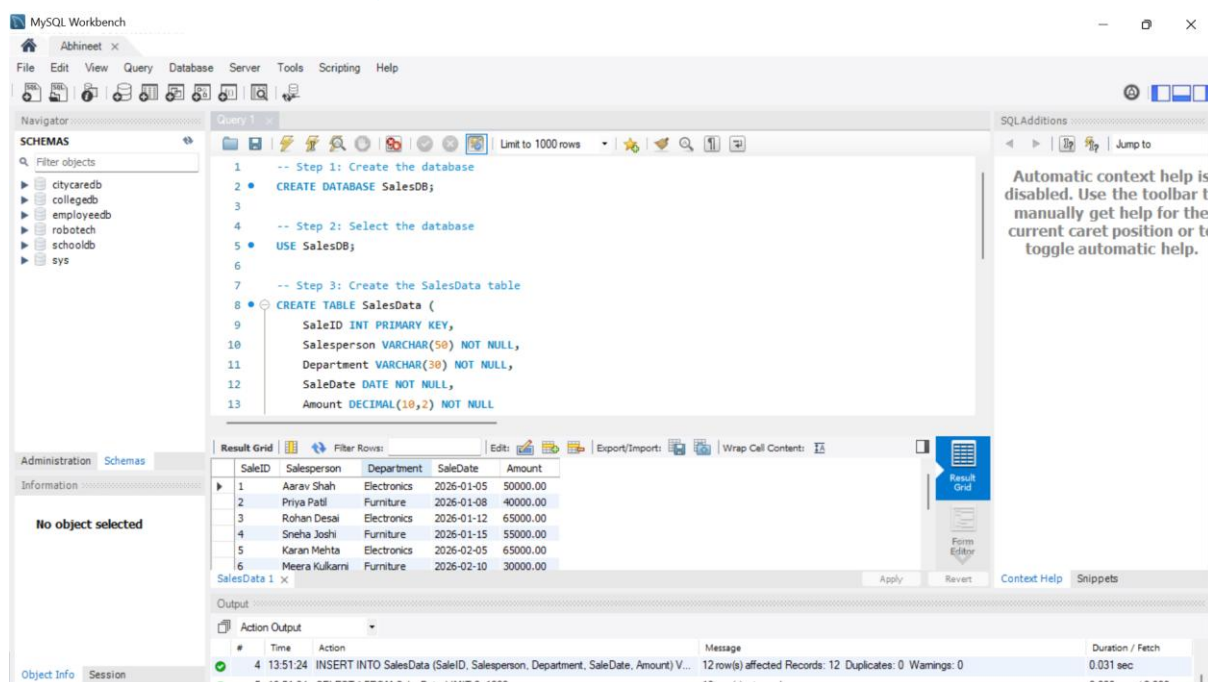
-- Step 4: Insert 12 sales records

```
INSERT INTO SalesData  
(SaleID, Salesperson, Department, SaleDate, Amount)  
VALUES  
(1, 'Aarav Shah', 'Electronics', '2026-01-05', 50000.00),  
(2, 'Priya Patil', 'Furniture', '2026-01-08', 40000.00),
```

```
(3, 'Rohan Desai', 'Electronics', '2026-01-12', 65000.00),
(4, 'Sneha Joshi', 'Furniture', '2026-01-15', 55000.00),
(5, 'Karan Mehta', 'Electronics', '2026-02-05', 65000.00),
(6, 'Meera Kulkarni', 'Furniture', '2026-02-10', 30000.00),
(7, 'Neha Shah', 'Clothing', '2026-02-12', 45000.00),
(8, 'Aman Verma', 'Clothing', '2026-02-18', 45000.00),
(9, 'Divya More', 'Electronics', '2026-03-03', 80000.00),
(10, 'Rahul Sharma', 'Furniture', '2026-03-06', 60000.00),
(11, 'Isha Nair', 'Clothing', '2026-03-10', 70000.00),
(12, 'Om Jadhav', 'Electronics', '2026-03-14', 55000.00);
```

-- Step 5: Display all records

SELECT * FROM SalesData;



Q2) Write a MySQL query using ROW_NUMBER() to assign a unique row number to every salesperson.

The result should be arranged by sale amount from highest to lowest.

Required Output

Column
Salesperson
Department
Amount
SalesRowNumber

```

USE SalesDB;

SELECT

    Salesperson,

    Department,

    Amount,

    ROW_NUMBER() OVER (

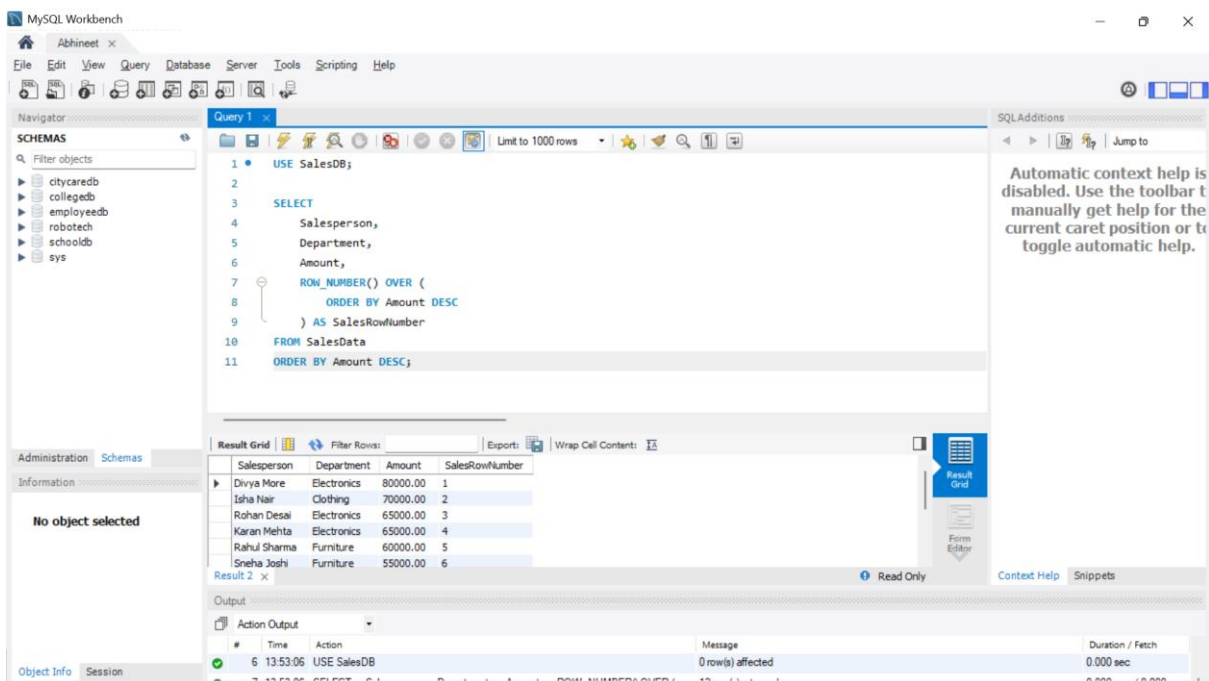
        ORDER BY Amount DESC

    ) AS SalesRowNumber

FROM SalesData

ORDER BY Amount DESC;

```



Q3) Write a MySQL query using RANK() to rank salespersons according to their sale amount.

Students having the same sale amount should receive the same rank.

Required Output

Column
Salesperson
Department
Amount

SalesRank

USE SalesDB;

SELECT

Salesperson,

Department,

Amount,

RANK() OVER (

ORDER BY Amount DESC

) AS SalesRank

FROM SalesData

ORDER BY Amount DESC;

The screenshot shows the MySQL Workbench interface. The query editor contains the following SQL code:

```
1 USE SalesDB;
2
3 SELECT
4     Salesperson,
5     Department,
6     Amount,
7     RANK() OVER (
8         ORDER BY Amount DESC
9     ) AS SalesRank
10 FROM SalesData
11 ORDER BY Amount DESC;
```

The Results window displays the following data:

Salesperson	Department	Amount	SalesRank
Divya More	Electronics	80000.00	1
Isha Nair	Clothing	70000.00	2
Rohan Desai	Electronics	65000.00	3
Karan Mehta	Electronics	65000.00	3
Rahul Sharma	Furniture	60000.00	5
Sneha Joshi	Furniture	55000.00	6

The Output window shows the following message:

#	Time	Action	Message	Duration / Fetch
10	13:55:04	USE SalesDB	0 row(s) affected	0.016 sec

Q4) Write a MySQL query to rank salespersons separately within each department.

Use:

Required Component

RANK()

OVER

PARTITION BY Department

ORDER BY Amount DESC

USE SalesDB;

SELECT

Salesperson,

Department,

Amount,

RANK() OVER (

PARTITION BY Department

ORDER BY Amount DESC

) AS DepartmentRank

FROM SalesData

ORDER BY Department, DepartmentRank;

The screenshot shows the MySQL Workbench interface. The query editor contains the following SQL query:

```
1 USE SalesDB;
2
3 SELECT
4     Salesperson,
5     Department,
6     Amount,
7     RANK() OVER (
8         PARTITION BY Department
9         ORDER BY Amount DESC
10    ) AS DepartmentRank
11 FROM SalesData
12 ORDER BY Department, DepartmentRank;
```

The Results tab shows the following data:

Salesperson	Department	Amount	DepartmentRank
Isha Nair	Clothing	70000.00	1
Neha Shah	Clothing	45000.00	2
Aman Verma	Clothing	45000.00	2
Divya More	Electronics	80000.00	1
Rohan Desai	Electronics	65000.00	2
Karan Mehta	Electronics	65000.00	2

The Output tab shows the following message:

#	Time	Action	Message	Duration / Fetch
12	13:56:02	USE SalesDB	0 row(s) affected	0.000 sec

Q5) Write a MySQL query to calculate the running total of sales according to the sale date.

Required Output

Column
SaleID
Salesperson
SaleDate
Amount
RunningTotal

USE SalesDB;

SELECT

SaleID,

Salesperson,

SaleDate,

Amount,

SUM(Amount) OVER (

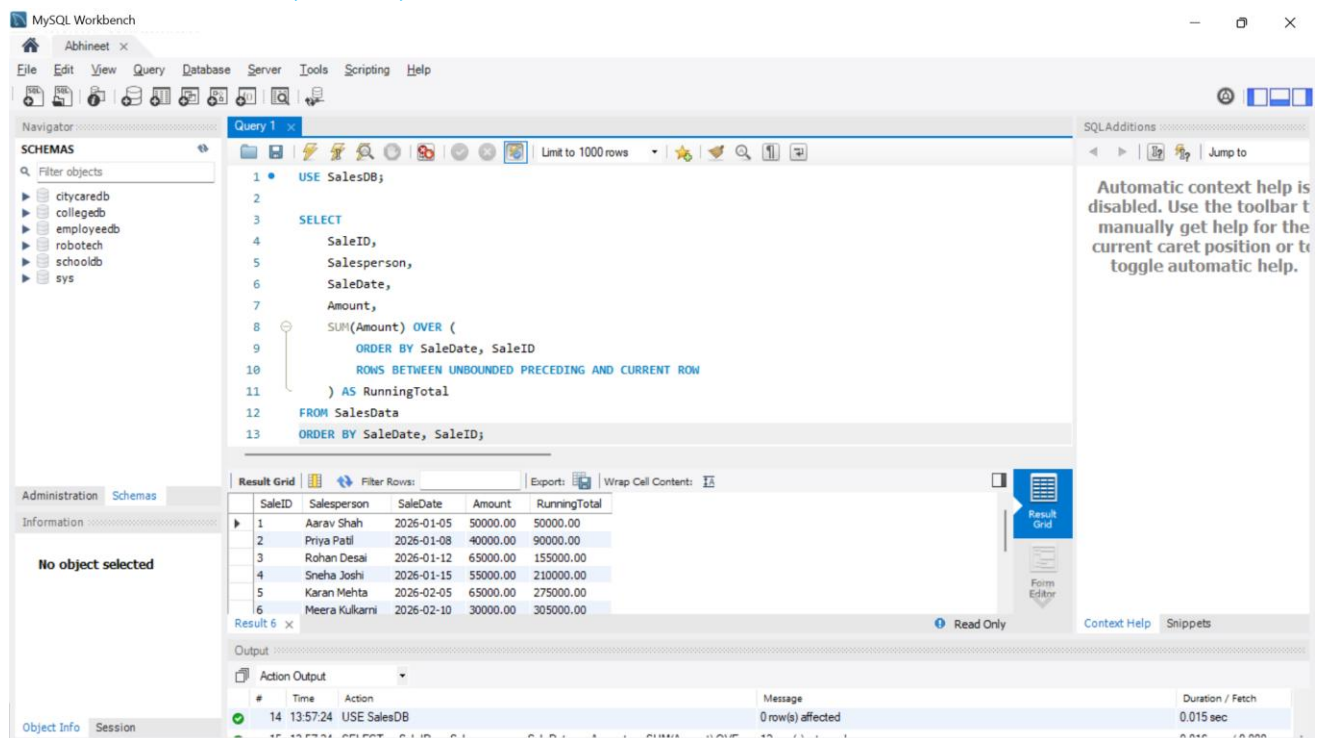
ORDER BY SaleDate, SaleID

ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW

) AS RunningTotal

FROM SalesData

ORDER BY SaleDate, SaleID;



Q6) Write a MySQL query to display the highest-performing salesperson from each department.

The query should:

Requirement

Use ROW_NUMBER()

Partition records by department

Arrange sale amounts in descending order

Use a subquery

Display only the first-ranked salesperson from every department

USE SalesDB;

SELECT

Salesperson,

Department,

Amount

FROM (

SELECT

Salesperson,

Department,

Amount,

ROW_NUMBER() OVER (

PARTITION BY Department

ORDER BY Amount DESC

) AS DepartmentRowNumber

FROM SalesData

) AS DepartmentSales

WHERE DepartmentRowNumber = 1;

The screenshot shows the MySQL Workbench interface. The main window displays a SQL query in the 'Query Editor' tab. The query is as follows:

```
USE SalesDB;

SELECT
    Salesperson,
    Department,
    Amount
FROM (
    SELECT
        Salesperson,
        Department,
        Amount,
        ROW_NUMBER() OVER (
            PARTITION BY Department
            ORDER BY Amount DESC
        ) AS DepartmentRowNumber
    FROM SalesData
) AS DepartmentSales
WHERE DepartmentRowNumber = 1;
```

The 'Result Grid' tab shows the results of the query. The results are as follows:

Salesperson	Department	Amount
Isha Nair	Clothing	70000.00
Divya More	Electronics	80000.00
Rahul Sharma	Furniture	60000.00

The 'Output' tab shows the execution of the query. The output is as follows:

#	Time	Action	Message	Duration / Fetch
16	13:59:08	USE SalesDB	0 row(s) affected	0.000 sec

